

CNN-Based CIFAR-10 Image Classification System Using Deep Learning

1. Problem Statement

Image classification is one of the most important applications of Deep Learning and Computer Vision. The objective of this project is to develop an intelligent image classification system capable of automatically identifying and categorizing images into predefined object classes using a Convolutional Neural Network (CNN). The system was trained on the CIFAR-10 dataset and integrated into a GUI-based application for real-time prediction.

2. Dataset Description

The CIFAR-10 dataset contains 60,000 RGB images divided into 10 classes. Each image has a resolution of 32×32 pixels. The dataset contains 50,000 training images and 10,000 testing images. The classes include airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

3. Preprocessing Techniques

Several preprocessing techniques were applied to improve model performance and stability. These include image normalization, data augmentation, random flipping, random rotation, random zooming, batch normalization, and dropout regularization. Image pixel values were normalized to the range [0,1] using rescaling.

4. Methodology

A custom CNN architecture was designed using multiple convolutional blocks. Each block consists of convolution layers, batch normalization, ReLU activation, max pooling, and dropout layers. The AdamW optimizer was used for efficient gradient optimization, while Sparse Categorical Crossentropy was selected as the loss function for multi-class classification. Several techniques were used to improve model performance:

- Data Augmentation → Improves generalization and reduces overfitting.
- Batch Normalization → Stabilizes training and accelerates convergence.
- Dropout → Prevents memorization and improves generalization.
- EarlyStopping → Stops training when validation loss stops improving.

- ReduceLROnPlateau → Dynamically reduces learning rate during training.

5. Training Configuration

Epochs: 50

Optimizer: AdamW

Learning Rate: $1e-3$

Loss Function: Sparse Categorical Crossentropy

Framework: TensorFlow / Keras

6. GUI-Based Application

A professional GUI-based application was developed using Streamlit. The interface allows users to upload an image and receive real-time predictions from the trained CNN model. The application also displays prediction confidence scores and graphical visualizations.

7. Results and Discussion

The model achieved strong performance on the CIFAR-10 dataset due to proper preprocessing, regularization, and optimization techniques. The use of data augmentation and dropout significantly reduced overfitting, while batch normalization improved training stability and convergence speed.

8. Conclusion

This project successfully implemented a CNN-based image classification system capable of accurately classifying CIFAR-10 images. The integration of a GUI-based application demonstrates the practical use of Artificial Intelligence in real-world computer vision systems.

9. Training Visualization

Figure 1: Epoch vs Accuracy

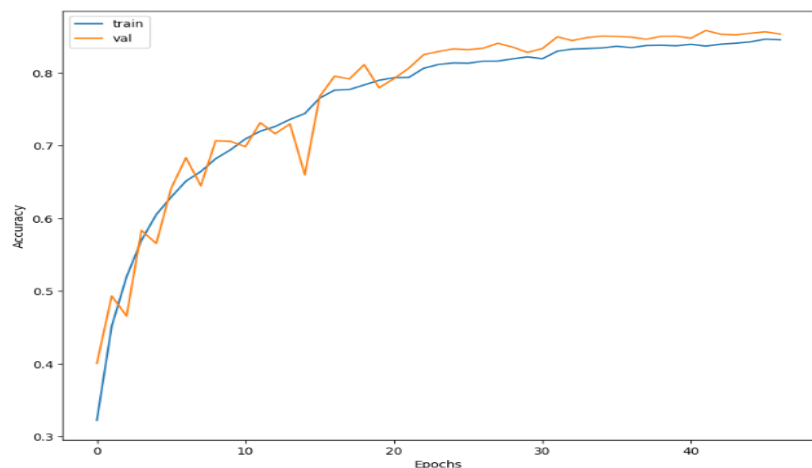
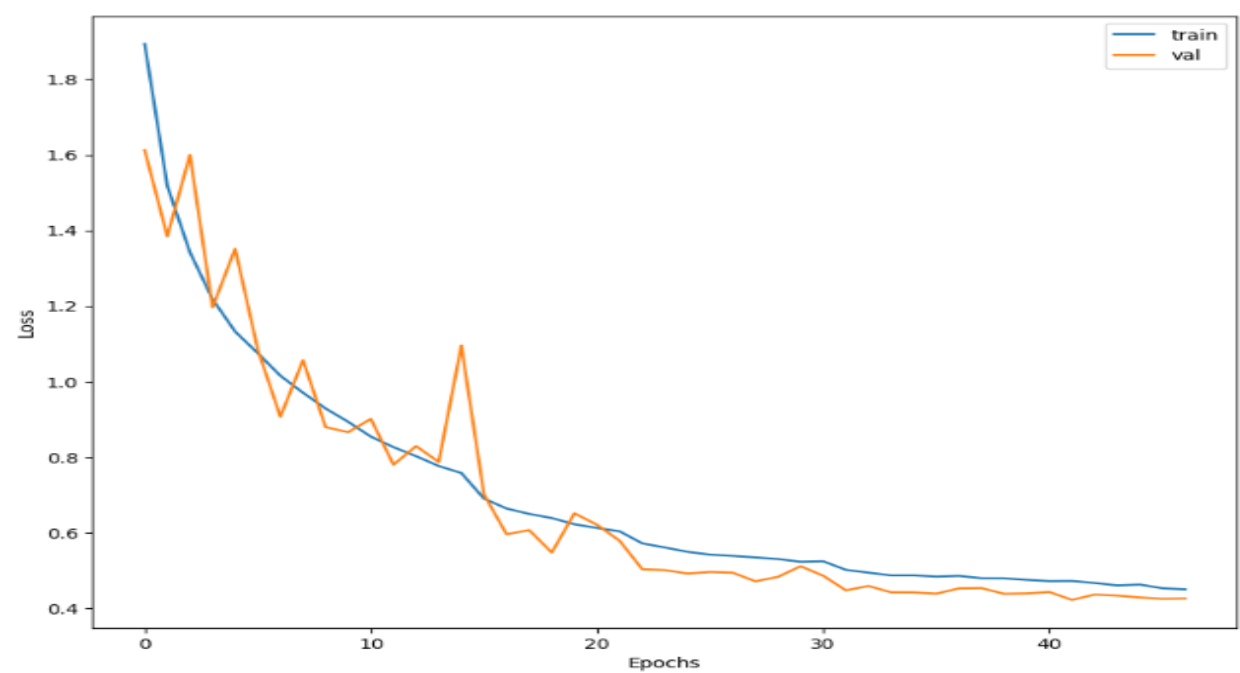


Figure 2: Epoch vs Loss



10. Classification Report

Metric	Description
Precision	Measures correct positive predictions
Recall	Measures ability to identify all positives
F1-Score	Harmonic mean of precision and recall
Accuracy	Overall prediction correctness

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